

LAB# 07

Introduction to the MySQL ORDER BY clause

```
SELECT
    Select list
FROM
    Table name
ORDER BY
    column1 [ASC|DESC],
    column2 [ASC|DESC],
    ...;
```

MySQL ORDER BY examples

We'll use the customers table from the sample database for the demonstration.

customers
* customerNumber
customerName
contactLastName
contactFirstName
phone
addressLine1
addressLine2
city
state
postalCode
country
salesRepEmployeeNumber
creditLimit

The following query uses the ORDER BY clause to sort the customers by their last names in ascending order.

```
SELECT
    contactLastname,
    contactFirstname
FROM
    customers
ORDER BY
    contactLastname;
```

```

+-----+-----+
| contactLastname | contactFirstname |
+-----+-----+
| Accorti         | Paolo           |
| Altagar,G M     | Raanan          |
| Andersen        | Mel             |
| Anton           | Carmen          |
| Ashworth        | Rachel          |
| Barajas         | Miguel          |
...

```

The following example uses the BETWEEN operator to find products whose buy prices between 90 and 100:

```

SELECT
    productCode,
    productName,
    buyPrice
FROM
    products
WHERE
    buyPrice BETWEEN 90 AND 100;

```

	productCode	productName	buyPrice
	S10_1949	1952 Alpine Renault 1300	98.58
	S10_4698	2003 Harley-Davidson Eagle Drag Bike	91.02
	S12_1099	1968 Ford Mustang	95.34
	S12_1108	2001 Ferrari Enzo	95.59
	S18_1984	1995 Honda Civic	93.89
	S18_4027	1970 Triumph Spitfire	91.92
	S24_3856	1956 Porsche 356A Coupe	98.3

The following example uses the IN operator to find the offices located in the USA and France:

```

SELECT
    officeCode,
    city,

```

```

    phone,
    country
FROM
    offices
WHERE
    country IN ('USA' , 'France');

```

```

+-----+-----+-----+-----+
| officeCode | city          | phone          | country |
+-----+-----+-----+-----+
| 1          | San Francisco | +1 650 219 4782 | USA    |
| 2          | Boston        | +1 215 837 0825 | USA    |
| 3          | NYC           | +1 212 555 3000 | USA    |
| 4          | Paris         | +33 14 723 4404 | France |
+-----+-----+-----+-----+
4 rows in set (0.01 sec)

```

This example uses the LIKE operator to find employees whose first names start with the letter a:

```

SELECT
    employeeNumber,
    lastName,
    firstName
FROM
    employees
WHERE
    firstName LIKE 'a%';

```

	employeeNumber	lastName	firstName
	1143	Bow	Anthony
	1611	Fixter	Andy

In MySQL, a NULL value means unknown. A NULL value is different from zero (0) or an empty string ". You can use a NULL value in the INSERT statement to specify that the data is missing. For example, the following statement inserts a row into the leads table. Because the phone number is missing, so a NULL value is used.

```

INSERT INTO leads(first_name,last_name,source,email,phone)
VALUE('John','Doe','Web Search','john.doe@acme.com',NULL);

```

Single row functions

Single row functions can be character functions, numeric functions, date functions, and conversion functions.

Case Conversion functions

The SELECT query below demonstrates the use of case conversion functions.

```
SELECT UPPER (first_name), INITCAP (last_name), LOWER (job_id)
FROM employees
WHERE rownum < 5;
```

UPPER(FIRST_NAME)	INITCAP(LAST_NAME)	LOWER(JOB_
STEVEN	King	ad_pres
NEENA	Kochhar	ad_vp
LEX	De Haan	ad_vp
ALEXANDER	Hunold	it_prog

Consider the following facts when using arithmetic operations in a SELECT statement:

1. There are seven arithmetic operators: Addition, Subtraction, Multiplication, Division, Modulo, DIV, Unary minus.

Operator	Description
+	Addition operator
-	Minus operator
*	Multiplication operator
/	Division operator
%	Modulo operator
DIV (v4.1.0)	Integer division
-	Unary minus. It changes the sign of the argument.

/*

Query 1: This query calculates the total price before discount and after discount for each order item.

*/

```
SELECT OrderID,
       ProductID,
       UnitPrice*Quantity AS "Regular Price",
       UnitPrice*Quantity-UnitPrice*Quantity*Discount AS "Price After Discount"
FROM order_details;
```

